Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Given the system of equations
$(b+c)(y+z)-ax=b-c$
$(c+a)(z+x)-by=c-a$
$(a+b)(x+y)-cz=a-b$
(where $a+b+c\neq 0$); then $x:y:z$ is given by

(A) $b-c:c-a:a-b$

(B) $b+c:c+a:a+b$

(C) $a:b:c$

(D) $\frac{a}{b}:\frac{b}{c}:\frac{c}{a}$

Q.2

Suppose the system of equations $a_1x+b_1y+c_1z=d_1$
$a_2x+b_2y+c_2z=d_2$
$a_3^2+b_3y+c_3^2z=d_3^2$
has a unique solution (x_0,y_0,z_0) . If $x_0=0$, then which one of the following is correct?

(A) $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}=0$

(B) $\begin{vmatrix} \mathrm{d}_1 & \mathrm{b}_1 & \mathrm{c}_1 \\ \mathrm{d}_2 & \mathrm{b}_2 & \mathrm{c}_2 \\ \mathrm{d}_3 & \mathrm{b}_3 & \mathrm{c}_3 \end{vmatrix}=0$

(C) $\begin{vmatrix} \mathrm{d}_1 & \mathrm{a}_1 & \mathrm{c}_1 \\ \mathrm{d}_2 & \mathrm{a}_2 & \mathrm{c}_2 \\ \mathrm{d}_3 & \mathrm{a}_3 & \mathrm{c}_3 \end{vmatrix}=0$

(D) $\begin{vmatrix} \mathrm{d}_1 & \mathrm{a}_1 & \mathrm{b}_1 \\ \mathrm{d}_2 & \mathrm{a}_2 & \mathrm{b}_2 \\ \mathrm{d}_3 & \mathrm{a}_3 & \mathrm{b}_3 \end{vmatrix}=0$

Q.3

The system of equations
$\alpha x+y+z=\alpha-1, x+\alpha y+z=\alpha-1, x+y+\alpha z=\alpha-1$
has no solution, if α is

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(A) not -2

(B) 1

(C) -2

(D) either -2 or 1

Q.4

If the system of linear equations:

$$x_1 + 2x_2 + 3x_3 = 6$$
$$x_1 + 3x_2 + 5x_3 = 9$$
$$2x_1 + 5x_2 + ax_3 = b$$

is consistent and has infinite number of solutions, then

(A) $a = 8, b$ can be any real number

(B) $b = 15, a$ can be any real number

(C) $a \in R - \{8\}$ and $b \in R - \{15\}$

(D) $a = 8, b = 15$

Q.5

For what value of p, is the system of equations :
$\mathrm{p}^3\mathrm{x} + (\mathrm{p}+1)^3\mathrm{y} = (\mathrm{p}+2)^3$
$px + (p+1)y = p + 2$
$x + y = 1$
consistent ?

(A) $\mathrm{p} = 0$

(B) $\mathrm{p} = 1$

(C) $\mathrm{p} = -1$

(D) For all $\mathrm{p} > 1$

Q.6

Consider the system of linear equations
$a_1x + b_1y + c_1z + d_1 = 0,$
$a_2x + b_2y + c_2z + d_2 = 0,$
$a_3x + b_3y + c_3z + d_3 = 0$
Let us denote by $\Delta(a, b, c)$ the determinant

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$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

If $\Delta(a,b,c)\#0$, then the value of x in the unique solution of the above equations is

(A) $\frac{\Delta(b,c,d)}{\Delta(a,b,c)}$

(B) $\frac{-\Delta(b,c,d)}{\Delta(a,b,c)}$

(C) $\frac{\Delta(a,c,d)}{\Delta(a,b,c)}$

(D) $-\frac{\Delta(a,b,d)}{\Delta(a,b,c)}$

Q.7

In a ΔABC, if $\begin{vmatrix} 1 & a & b \\ 1 & c & a \\ 1 & b & c \end{vmatrix} = 0$, then value of $\tan^2\frac{A}{2} + \tan^2\frac{B}{2} + \tan^2\frac{C}{2}$ is equal to

[Note: All symbols used have usual meaning in ΔABC.]

(A) -1

(B) 0

(C) 1

(D) $\frac{1}{3}$

Q.8

Let λ and α be real. The set of all values of λ for which the system of linear equations $\lambda x + (\sin\alpha)y + (\cos\alpha)z = 0$
$x + (\cos\alpha)y + (\sin\alpha)z = 0$
$-x + (\sin\alpha) - (\cos\alpha)z = 0$ has a non-trivial
solution, is

(A) $[0, \sqrt{2}]$

(B) $[-\sqrt{2}, 0]$

(C) $[-\sqrt{2}, \sqrt{2}]$

(D) None of these

Q.9

For what values of k, does the system of linear equation $x + y + z = 2, 2x + y - z = 3$,
$3x + 2y + kz = 4$
have a unique solution ?

(A) $\mathrm{k} = 0$

(B) $-1 < \mathrm{k} < 1$

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(C) $-2 < \mathrm{k} < 2$

(D) $\mathrm{k} \neq 0$

Q.10

The number of values of k for which the system of equations $(k+1)x+8y=4k; kx+(k+3)y=3k-1$ has infinitely many solutions is

(A) 0

(B) 1

(C) 2

(D) infinite

Q.11

If the system of linear equations
$x+2ay+az=0; x+3by+bz=0;$
$x+4cy+cz=0$ has a non - zero solution, then $\mathrm{a}, \mathrm{b}, \mathrm{c}$.

(A) satisfy $a+2b+3c=0$

(B) are in A.P

(C) are in G..P

(D) are in H.P.

Q.12

If the system of equations $\lambda x_1+x_2+x_3=1\, \mathrm{x}_1+\lambda \mathrm{x}_2+\mathrm{x}_3=1, \mathrm{x}_1+\mathrm{x}_2+\lambda \mathrm{x}_3=1$ is consistent then λ can be

(A) 5

(B) $-2/3$

(C) -3

(D) None of these

Q.13

The equations $2x+3y+4=0; 3x+4y+6=0$ and $4x+5y+8=0$ are

(A) consistent with unique solution

(B) inconsistent

(C) consistent with infinitely many solutions

(D) None of the above

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Q.14

Given $\mathrm{a} = \mathrm{x}/(\mathrm{y} - \mathrm{z}), \mathrm{b} = \mathrm{y}/(\mathrm{z} - \mathrm{x})$ and $\mathrm{c} = \mathrm{z}/(\mathrm{x} - \mathrm{y})$
where x, y and z are not all zero, Then the value of $\mathrm{ab} + \mathrm{bc} + \mathrm{ca}$

(A) 0

(B) 1

(C) -1

(D) None of these

Q.15

Let a, b, c be any real numbers. Suppose that there are real numbers x, y, z not all zero such that $x = cy + bz, y = az + cx$ and $z = bx +$ ay. Then $a^2 + b^2 + c^2 + 2abc$ is equal to

(A) -1

(B) 0

(C) 1

(D) 2

Answers & Solutions

Q.1 Answer:

$b - c : c - a : a - b$

Solution:

$(b + c)(y + z) - ax = b - c$ ... **(1)**
$(c + a)(z + x) - by = c - a$ ... **(2)**
$(a + b)(x + y) - cz = a - b$

Adding all three equation

$(x + y + z)(a + b + c) = 0$
$\Rightarrow \mathrm{x} + \mathrm{y} + \mathrm{z} = 0$
$\Rightarrow \mathrm{y} + \mathrm{z} = -\mathrm{x}$

Substituting this in the first (1) equation

$(b + c)(-x) - ax = b - c$
$x = \frac{c-b}{a+b+c}$

Similarly, we get

$y = \frac{a-c}{a+b+c}, z = \frac{b-a}{a+b+c}$
Hence $\mathrm{x} : \mathrm{y} : \mathrm{z} = (\mathrm{c} - \mathrm{b}) : (\mathrm{a} - \mathrm{c}) : (\mathrm{b} - \mathrm{a})$

Q.2 Answer:

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$$\begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix} = 0$$

Solution:

The given system of equations is $a_1x + b_1y + c_1z = d_1$

$a_2x + b_2y + c_2z = d_2$

and $a_3x + b_3y + c_3z = d_3$

Let $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

This system has a unique solution x_0, y_0, z_0

if $\Delta \neq 0$ **and** $x_0 = \frac{\Delta x}{\Delta} = 0 \Rightarrow \Delta x = 0$

$$\Rightarrow \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix} = 0$$

Q.3 Answer:

-2

Solution:

The value of the coefficient determinant

$D = \begin{vmatrix} \alpha & 1 & 1 \\ 1 & \alpha & 1 \\ 1 & 1 & \alpha \end{vmatrix} = 0$ **which gives** $(\alpha - 1)^2[\alpha + 2] = 0$. **So,**

$\alpha = 1$ **or** $\alpha = -2$ **but for** $\alpha = 1$ **there are infinite number of solutions of the given system.**

Q.4 Answer:

$a = 8, b = 15$

Solution:

Given system of equations can be written in matrix form as $AX = B$ **where**

$A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 2 & 5 & a \end{pmatrix}$ **and** $B = \begin{pmatrix} 6 \\ 9 \\ b \end{pmatrix}$

Since, system is consistent and has infinitely many solutions $\therefore$ **(adj.** $A)B = 0$

$$\Rightarrow \begin{pmatrix} 3a - 25 & 15 - 2a & 1 \\ 10 - a & a - 6 & -2 \\ -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 6 \\ 9 \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$\Rightarrow -6 - 9 + b = 0 \Rightarrow b = 15$

and $6(10 - a) + 9(a - 6) - 2(b) = 0$

$\Rightarrow 60 - 6a + 9a - 54 - 30 = 0$

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$\Rightarrow 3a = 24 \Rightarrow a = 8$

Hence, $a = 8, b = 15.$

Q.5 Answer:

$\mathrm{p} = -1$

Solution:

The given system of equations are :

$\mathrm{p}^3\mathrm{x} + (\mathrm{p}+1)^3\mathrm{y} = (\mathrm{p}+2)^{\frac{3}{3}} \quad \ldots$

$px + (p+1)y = (p+2)$

$x + y = 1$

This system is consistent, if values of x **and** y **from first two equation satisfy the third equation.**

which $\Rightarrow \begin{vmatrix} p^3 & (p+1)^3 & (p+2)^3 \\ p & (p+1) & (p+2) \\ 1 & 1 & 1 \end{vmatrix} = 0$

$\Rightarrow \begin{vmatrix} p^3 & (p+1)^3 - p^3 & (p+2)^3 - p^3 \\ p & 1 & 2 \\ 1 & 0 & 0 \end{vmatrix} = 0$

$\Rightarrow 2(\mathrm{p}+1)^3 - 2\mathrm{p}^3 - (\mathrm{p}+2)^3 + \mathrm{p}^3 = 0$

$\Rightarrow 2\left(\mathrm{p}^3 + 1 + 3\mathrm{p}^2 + 3\mathrm{p}\right) - 2\mathrm{p}^3 - \left(\mathrm{p}^3 + 8 + 12\mathrm{p} + 6\mathrm{p}^2\right) + \mathrm{p}^3 = 0$

$\Rightarrow 2p^3 + 2 + 6p^2 + 6p - 2p^3 - p^3 - 8 - 12p - 6p^2 + p^3 = 0$

$\Rightarrow -6 - 6p = 0$

$\Rightarrow \quad p = -1$

Q.6 Answer:

$\frac{\Delta(b,c,d)}{\Delta(a,b,c)}$

Solution:

From the given system of equations,

$x = \frac{D_1}{D}, y = \frac{D_2}{D}, z = \frac{D_3}{D}$ **where,**

$D = \Delta(a,b,c)$

$D_1 = \Delta(d,b,c)$

$D_2 = \Delta(a,d,c)$

$D_3 = \Delta(a,b,d)$

Now, $x = \frac{\Delta(d,b,c)}{\Delta(a,b,c)}$

where, $\Delta(d,b,c) = \begin{vmatrix} -d_1 & b_1 & c_1 \\ -d_2 & b_2 & c_2 \\ -d_3 & b_3 & c_3 \end{vmatrix}$

$= -\begin{vmatrix} b_1 & -d_1 & c_1 \\ b_2 & -d_2 & c_2 \\ b_3 & -d_3 & c_3 \end{vmatrix} = +\begin{vmatrix} b_1 & c_1 & -d_1 \\ b_2 & c_2 & -d_2 \\ b_3 & c_3 & -d_3 \end{vmatrix}$

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$= \Delta(b, c, d)$

Hence, $x = \frac{\Delta(b,c,d)}{\Delta(a,b,c)}$

Q.7 Answer:

1

Solution:

$$\begin{vmatrix} 1 & a & b \\ 1 & c & a \\ 1 & b & c \end{vmatrix} = 0$$

$R_2 \to R_2 - R_1, \quad R_3 \to R_3 - R_2$

$$\begin{vmatrix} 1 & a & b \\ 0 & c-a & a-b \\ 0 & b-c & c-a \end{vmatrix} = 0$$

$\Rightarrow 1(c-a)^2 - a - bb - c = 0$

$\Rightarrow c^2 + a^2 - 2ca - ab + ac + b^2 - bc = 0$

$\Rightarrow a^2 + b^2 + c^2 - ab - bc - ca = 0$

$\Rightarrow \frac{1}{2}a - b^2 + b - c^2 + c - a^2 = 0$

$\therefore a = b, \quad b = c, \quad c = a \Rightarrow a = b = c$

$\therefore \quad \angle A = \angle B = \angle C = 60°$

$\therefore \quad \tan^2\frac{A}{2} + \tan^2\frac{B}{2} + \tan^2\frac{C}{2}$

$= \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$

Q.8 Answer:

$[-\sqrt{2}, \sqrt{2}]$

Solution:

Since the system has a non-trivial solution,

therefore $\begin{vmatrix} \lambda & \sin\alpha & \cos\alpha \\ 1 & \cos\alpha & \sin\alpha \\ -1 & \sin\alpha & -\cos\alpha \end{vmatrix} = 0$

$\Rightarrow \lambda\left(-\cos^2\alpha - \sin^2\alpha\right)$

$-(-\sin\alpha\cos\alpha - \sin\alpha\cos\alpha)$

$-\left(\sin^2\alpha - \cos^2\alpha\right) = 0$

$\Rightarrow -\lambda + \sin 2\alpha + \cos 2\alpha = 0$

$\Rightarrow \lambda = \sin 2\alpha + \cos 2\alpha$

$\Rightarrow \lambda = \sqrt{2}\cos\left(2\alpha - \frac{\pi}{4}\right)$

Since $-1 \le \cos\left(2\alpha - \frac{\pi}{4}\right) \le 1 \forall \in \mathrm{R}$

$\therefore \quad -\sqrt{2} \le \lambda \le \sqrt{2}$ **i.e.** $\lambda \in [-\sqrt{2}, \sqrt{2}]$

Q.9 Answer:

$k \neq 0$

Solution:

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The given system of equations is $x + y + z = 2$

...(i) $2x + y - z = 3$

and $3x + 2y + kz = 4 \quad \ldots$ **(iii)**

This system has a unique solution if $\begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 3 & 2 & \mathrm{k} \end{vmatrix} \neq 0$

Applying $\mathrm{C}_2 \to \mathrm{C}_2 - \mathrm{C}_1$ **and** $\mathrm{C}_3 \to \mathrm{C}_3 - \mathrm{C}_1$

We get $\begin{vmatrix} 1 & 0 & 0 \\ 3 & -1 & -3 \\ 3 & -1 & \mathrm{k} - 3 \end{vmatrix} \neq 0$

$\Rightarrow -1(\mathrm{k} - 3) - 3 \neq 0$ **or** $-\mathrm{k} + 3 - 3 \neq 0 \Rightarrow \mathrm{k} \neq 0$

Q.10 Answer:

1

Solution:

For infinitely many solutions the two equations become identical $\Rightarrow \quad \frac{k+1}{k} = \frac{8}{k+3} = \frac{4k}{3k-1} \Rightarrow k = 1$

Q.11 Answer:

are in H.P.

Solution:

For homogeneous system of equations to have non zero solution, $\Delta = 0$

$\begin{vmatrix} 1 & 2a & a \\ 1 & 3b & b \\ 1 & 4c & c \end{vmatrix} = 0 \left[\because C_2 \to C_2 - 2C_3\right]$

$\begin{vmatrix} 1 & 0 & a \\ 1 & b & b \\ 1 & 2c & c \end{vmatrix} = 0 \left[R_3 \to R_3 - R_2, R_2 \to R_2 - R_1\right]$

On simplification, $\frac{2}{b} = \frac{1}{a} + \frac{1}{c}$

$\therefore a, b, c$ **are in Harmonic Progression.**

Q.12 Answer:

None of these

Solution:

Let $\Delta = \begin{vmatrix} \lambda & 1 & 1 \\ 1 & \lambda & 1 \\ 1 & 1 & \lambda \end{vmatrix} = \begin{vmatrix} \lambda + 2 & 1 & 1 \\ \lambda + 2 & \lambda & 1 \\ \lambda + 2 & 1 & \lambda \end{vmatrix}$

$[\mathrm{C}_1 \to \mathrm{C}_1 + \mathrm{C}_2 + \mathrm{C}_3]$

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$= (\lambda+2)\begin{vmatrix} 1 & 1 & 1 \\ 1 & \lambda & 1 \\ 1 & 1 & \lambda \end{vmatrix} = (\lambda+2)\begin{vmatrix} 1 & 0 & 0 \\ 1 & \lambda-1 & 0 \\ 1 & 0 & \lambda-1 \end{vmatrix}$

$= (\lambda+2)(\lambda-1)^2$

$[$ **using** $C_2 \to C_2 - C_1$ **and** $C_3 \to C_3 - C_1]$

But when $\lambda = 1$, the system of equation becomes $x_1 + x_2 + x_3 = 1$ which has infinite number of solutions. When $\lambda = -2$, by adding three equations, we obtain $0 = 3$ and thus, the system of equations is inconsistent.

Q.13 Answer:

consistent with unique solution

Solution:

Consider first two equations:

$2x + 3y = -4$ **and** $3x + 4y = -6$

We have $\Delta = \begin{vmatrix} 2 & 3 \\ 3 & 4 \end{vmatrix} = -1 \neq 0$

$\Delta_x = \begin{vmatrix} -4 & 3 \\ -6 & 4 \end{vmatrix} = 2$ **and** $\Delta_y = \begin{vmatrix} 2 & -4 \\ 3 & -6 \end{vmatrix} = 0$

$\therefore \quad x = -2$ **and** $y = 0$

Now this solution satisfies the third, so the equations are consistent with unique solution.

Q.14 Answer:

-1

Solution:

$a = x/(y-z) \Rightarrow x - ay + az = 0$

$b = y/(z-x) \Rightarrow bx + y - bz = 0 \quad \ldots(2)$

$c = z/(x-y) \Rightarrow -cx + cy + z = 0 \ldots(3)$

As x, y, z are not all zero, the above system has non-trivial

solution so $\Delta = \begin{vmatrix} 1 & -a & a \\ b & -1 & -b \\ -c & c & 1 \end{vmatrix}$

$1 + ab + bc + ca = 0$

Q.15 Answer:

1

Solution:

$x - cy - bz = 0$

$cx - y + az = 0$

$bx + ay - z = 0$

for non zero solutions, $|A| = 0$

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$$\begin{vmatrix} 1 & -c & -b \\ c & -1 & a \\ b & a & -1 \end{vmatrix} = 0$$

$\Rightarrow \quad 1\left(1-a^2\right)+c(-c-ab)-b(ac+b)=0$

$\Rightarrow \quad 1-a^2-c^2-abc-b^2-abc=0$

$\Rightarrow a^2+b^2+c^2+2abc=1$

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